How a reported number is produced

From the frozen specification to the macro the prose cites, with no hand-copied value anywhere on the path

Frozen specification

spec/SPECIFICATION.md fixes the API, state machine, data structures, and concurrency model, before any code is written.

Four build variants

Two languages crossed with two allocators. Matched optimisation levels. Both sides pin the same shard and intra-shard hashes.

Deterministic driver

A fixed SplitMix64 draw order and a per-slot expected-state model, so both versions walk identical code paths.

Service and capture

Eight shards, one owning thread each, 1024-slot queues. HDR histogram timed from the intended send time, not the actual.

Repeated 20 times, with the four variants interleaved inside each repeat

14 offered-load levels up to 784 kops/s give 1120 measured runs. Interleaving makes the interference common to a block, so it cancels.

Raw measurement logs

bench/results/*.jsonl, with the environment record from capture_env.sh kept alongside each result set.

bench/plot_figures.py

Median with a bootstrap 95% confidence interval, an exact paired permutation test, and the Holm correction.

Generated artifacts

figures/*.pdf
figures/tab_*.tex
figures/gen_macros.tex

Thesis LaTeX source

The prose refers to the generated macros by name, so text and data cannot drift.

Nothing in the evaluation is hand-copied

bench/refresh_thesis.sh reruns the analysis and rebuilds the PDF, so a fresh campaign updates every figure, table, and quoted number in one step.

A third party can walk this path without proprietary code: the specification and the harness are themselves published artifacts.